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Research Paper

Recreational visits to urban parks and factors affecting park visits: Evidence from geotagged social media data

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ABSTRACT

Quantifying park use and understanding its driving factors is crucially important for increasing park use and thus human well-being. Previous studies have investigated the effects of different physical and sociocultural factors on park usage using visitor surveys and direct observations of park users, which are usually site specific and time consuming. We quantified and compared the number of visits for different types of parks in Beijing using freely available geotagged check-in data from social media. We investigated how park attributes, park location, park context and public transportation affected the number of park check-in visits, using multiple linear regressions. Despite potential biases in the use of social media data, using a park typology, we found that the number of visits was significantly different among different types of parks. While cultural relics parks and large urban parks had larger numbers of visits, neighborhood parks had higher visitation rates per unit of area. Park size and entrance fees were associated with increased numbers of visits for all types of parks. For parks that mainly serve local residents, the distance to urban center significantly affected park use. The number of bus stops was positively correlated with park visits, suggesting that increased accessibility through public transportation leads to more visits. The results indicated that improving park accessibility via public transportation and planning small, accessible green spaces in residential areas were effective in improving park use.

1. Introduction

Urban greenspaces, particularly parks, provide multiple benefits to human well-being, including physical and psychological health and social benefits (Hand et al., 2017; Huang & Cadenasso, 2016; Richardson, Pearce, Mitchell, & Kingham, 2013; Tzoulas et al., 2007; Van Herzele & de Vries, 2011; Wright Wendel, Zarger, & Mihelcic, 2012; Zhou, Wang, & Cadenasso, 2017). Park visits offer opportunities to directly experience the benefits of "natural" ecosystems, particularly for urban residents who have limited contact with natural environments (Daniel et al., 2012). Therefore, quantifying residents' visits to urban parks and understanding the factors that influence their visits are crucially important for the planning and management of urban parks.

Quantifying visits to parks and urban greenspaces is essential to understanding their recreational values, and identifying the influencing factors. Traditional methods of measuring the number of visitors include visitor surveys, direct observation and on-site counters (Cohen et al., 2010; Giles-Corti et al., 2005; Wright Wendel et al., 2012). Such methods of systematic observation typically select a representative sample of urban parks, and collect park usage information, such as the number of visits and the characteristics, activities and behavior of park users. Such methods, however, are usually site specific and time consuming, and thus have limited spatial coverage (Sessions, Wood, Rabotyagov, & Fisher, 2016; Tenkanen et al., 2017).

In addition to traditional survey methods, public participation geographic information systems (PPGIS) have increasingly been employed to measure recreational use of parks and to engage the public in generating spatially explicit information for planning purposes (Brown, Schebella, & Weber, 2014; Laatikainen, Tenkanen, Kyttä, & Toivonen, 2015; Wolf, Wohlfart, Brown, & Bartolomé Lasa, 2015). The PPGIS approach recruits participants to identify the locations where they engage in physical activities (Brown et al., 2014). Online PPGIS mapping has increasingly been used to enhance efficiency compared with field-based PPGIS mapping (Wolf et al., 2015). Additionally, PPGIS has been improved by using actual GPS tracking-based visitor distributions (versus those reported in PPGIS). GPS tracking has been greatly advanced by using participants' own tracking devices (e.g., smartphones) rather than devices supplied by researchers, making participation

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accessible to anyone owning a smartphone (Wolf et al., 2015; Wolf, Hagenloh, & Croft, 2012; Wolf, Stricker, & Hagenloh, 2013).

The emergence of freely available social media data provides new approaches for measuring visits to urban parks. Information from social media and other sources of “big data” increases in volume each year, and it can be used to study how people interact with real environments and to evaluate user preferences across space and time (Wood, Guerry, Silver, & Lacayo, 2013). As opposed to PPGIS and GPS tracking which are characterized by purposeful sampling and a structured data collection process, social media data are not generated for the explicit purpose of being used in specific research projects. Such types of data have been labeled with many terms, such as volunteered geographic information (VGI), citizen sensors, user-generated content (UGC) and crowdsourced geodata (Dunkel, 2015). This type of data has been used on a large scale to measure people’s preferences for various natural environments, such as lakes (Keeler et al., 2015), protected areas (Levin, Kark, & Crandall, 2015; Tenkanen et al., 2017), national parks (Heikinheimo et al., 2017; Sessions et al., 2016), mountain landscapes (Tenerelli, Demšar, & Luque, 2016), natural treatment systems (Ghermandi, 2016), ecologically engineered wetlands (Ghermandi, 2017), and urban greenway networks (Liu, Siu, Gong, Gao, & Lu, 2016). Data from web-share services have been used to monitor recreational activities in protected areas and urban parks (Campelo & Nogueira Mendes, 2016; Nogueira Mendes, Dias, & Pereira da Silva, 2014; Nogueira Mendes, Silva, Grilo, Rosalino, & Silva, 2012; Santos, Nogueira Mendes, & Vasco, 2014, 2016). In addition, geotagged photos have been used to estimate the aesthetic value in Hokkaido (Yoshimura & Hiura, 2017) and to reveal multiple landscape values around hydroelectric dams and reservoirs (Chen, Parkins, & Sherren, 2017). Most of the emerging VGI studies were conducted in developed western countries, using data from Instagram, Twitter or Flickr. However, few studies have been conducted using data from popular social media platforms to estimate visitation patterns in China (Tenkanen et al., 2017).

A considerable amount of research has investigated the effects of different physical and sociocultural factors on park usage (Giles-Corti et al., 2005; Grow et al., 2008; Mowen, Orsega-Smith, Payne, Ainsworth, & Godbey, 2007; Zhang, Chen, Sun, & Bao, 2013; Zhang, Yang, Ma, & Huang, 2015). For example, several studies have found that distance to parks, park size and park attractiveness (e.g., shade along paths and the presence of sports facilities, water features and birds) influence residents’ likelihood to use public parks (Giles-Corti et al., 2005). The quality of vegetation is correlated with the recreational appropriateness of urban parks (Zhang et al., 2013, 2015). The population density around parks also plays an important role because people who live closer to a park are more likely to visit it (Mowen et al., 2007). At the same time, the socioeconomic and sociodemographic characteristics of park users also influence park use (Jim & Shan, 2013; Sanesi & Chiarello, 2006; Schipperijn, Stigsdotter, Randrup, & Troelsen, 2010; Wright Wendel et al., 2012). Other variables affecting park use include users’ perceptions of safety (Jansson, Fors, Lindgren, & Wiström, 2013; Wang, Brown, & Liu, 2015), the provision of park facilities (Sugiyama & Ward Thompson, 2008; Wright Wendel et al., 2012), and the number of organized activities (Cohen et al., 2010).

These studies have advanced the understanding of factors that influence park use. However, these studies have typically been conducted using visitor surveys and/or interviews and direct observations of park users and activities; these are usually site-specific and time-consuming and include only small groups of people. Recently, however, a small number of studies have used the increasingly available and ready-to-use VGI data, especially in developing countries such as China. While quantifying urban park visits is important for measuring the recreational use of these sites, research on urban park visitation is extremely limited in Chinese cities due to the lack of data on residents’ actual park visits (Liu, Li, Xu, & Han, 2017). Consequently, the needs of residents have often been overlooked in the park construction and management process in China (Zhang & Yang, 2014). With increasingly available VGI data, which can potentially provide a proxy for the number of park visits, it is hoped that research using VGI data can fill this gap. Here, we use geotagged Weibo check-in data, a type of VGI data used as a measure (or proxy) of park visits, to identify and map visits to different types of parks in central Beijing. Specifically, the objectives are to 1) quantify park visits and characterize park visiting intensity among different types of parks and 2) examine the spatial-physical and socioeconomic factors that affect visits to urban park. The results from this study can provide important insights into urban park management and planning.

2. Methods

2.1. Study area

The study was conducted in Beijing, the capital city of China. Beijing has an administrative area of 16,808 km^2. The latest census, from 2016, shows that Beijing had 21.73 million permanent residents. Beijing had 868.09 km^2 of green coverage, with a total area of park green spaces of 295.03 km^2 and per capita park green space of 22 m^2 in 2015 (Beijing Municipal Bureau of Landscape and Forestry, 2016). There are 410 parks of various types in Beijing (Beijing Municipal Bureau of Landscape and Forestry, 2017), in addition to numerous small parks that are not counted, but also contribute to the urban greenspace system.

We focused on the 127 parks within the 5th ring road of Beijing, an urban core with an area of approximately 660 km^2, where check-in records were available. These include 91 parks listed on the official website of the Beijing Municipal Bureau of Landscape and Forestry, as well as 36 small parks that are not listed. The boundaries of the 127 parks were drawn based on Google Earth imagery. We omitted the 16 small parks within the 5th ring road that had zero check-ins.

2.2. Number of park visits based on Weibo check-in data

We used check-in data from Sina Weibo to measure the numbers of visits to urban parks. Weibo is the Chinese equivalent of Twitter and is the largest social media site in China. Weibo has a large number of users and represents one of the largest available geotagged datasets. According to Weibo’s annual report, it had 132 million daily active users in 2016, accounting for more than 9% of the Chinese population. Of Weibo users, 77.8% has high levels of education (college degrees or above), and there are more male users than female users (Weibo Data Center, 2016). Therefore, the demographic characteristics of Weibo users are not consistent with the total population. The public interfaces of Weibo’s location-based services (LBS) were launched on May 28, 2012. From that time, Weibo users could share their real-time locations on the Internet. As one type of fine-scaled open crowdsourcing data, the Weibo check-in data were the most appropriate dataset we could obtain as a proxy to estimate actual visits to parks. It is unfortunate that we were not able to obtain real park visit data to validate its relationship with the Weibo check-in data. However, a previous study focusing on 87 urban parks in Shanghai, China, has shown that there was a significant correlation between check-in data from the social media platform Jiepang and official visitor statistics (Shen, Sun, & Che, 2017). Additionally, previous studies have shown that Weibo check-in data can well reflect people’s preferences and activities in urban spaces. Indexes such as activity closeness, connection and intensity were calculated based on Weibo check-in data at the regional scale to delineate urban boundaries (Zhen, Cao, Qin, & Wang, 2017). Using Weibo check-in data, Shen and Karimi (2016) proposed a novel model that integrates multiple dimensions of the urban function network and thereby enriches the description of the urban network system. Furthermore, although using Weibo check-ins as a proxy of visitation is still rare, previous studies that used data from similar social media platforms, such as Instagram, Twitter and Flickr, have found significant positive relationships between official visitor statistics and visitation numbers estimated by these platforms (Keeler et al., 2015; Tenkanen et al., 2017; Wood et al., 2013).

Data collection was facilitated by Weibo application program interfaces (APIs). Through the “to obtain nearby locations” API, we retrieved identifications of park-related locations (e.g., “park”, “public square”, “outdoors”, “scenic spot”, “park entrance”, “zoo”, and “garden”) within the central urban area of Beijing. Thereafter, using the “to obtain location details” API, we obtained the latitudes, longitudes, names and check-in numbers of the retrieved locations. Weibo APIs allow a maximum number of 1000 calls per hour, which can well fulfill the data collection process in our study. Data were retrieved from Weibo on August 31, 2016. The check-in data covered a 4-year time period from May 2012 to August 2016.

After obtaining this location information, we found that some locations we obtained were not parks (for example, pedestrian walkways, the former residences of celebrities, and sculptures). We examined the locations one by one and deleted those that were not within the park category. Afterwards, for all parks with check-in locations, the total number of Weibo check-ins within each park was calculated. After processing, our data returned a total of 581,354 geotagged check-in visits for parks in the central urban area. Check-in data collection was conducted using Python Programming Language (version 2.7.12).

2.3. Park type classification

Parks in Beijing are classified into six mutually exclusive categories: (1) recreational parks (n = 13), (2) cultural relics parks (n = 16), (3) large urban parks (n = 8), (4) natural parks (n = 14), (5) community parks (n = 31), and (6) neighborhood parks (n = 45) (Table 1). These parks were classified based on their dominant functions and types of management (Tao, Chen, Zhang, & Bai, 2013). Fig. 1 shows the distribution of different types of urban parks within the study area. Among the 127 parks, the most popular type was the neighborhood park, followed by the community park. Both types were evenly distributed within the 5th ring road. Natural parks were located largely between the 4th and 5th ring roads. Cultural relics parks and large urban parks were mainly distributed in the central and northern parts of the study area. Recreational parks were found mostly in the western part.

2.4. Statistical analysis

We first compared the differences in total park check-in visits and visits per unit of park area (referred to hereafter as visiting intensity) among the six types of parks, assuming that total park check-in visits would be different from visiting intensity. We used the Kruskal-Wallis test to compare differences in park visits and visiting intensity among park types, and a Dunn-Bonferroni test for pairwise multiple comparisons. We used the nonparametric Kruskal-Wallis test instead of the analysis of variation approach (ANOVA) because the check-in data were not normally distributed.

We then used Spearman’s correlation to analyze the bivariate correlations between park check-in visits and their potential influencing factors. These included several characteristics of the parks and their surrounding areas, park accessibility, and the spatial locations of parks. Specifically, we selected 9 predictor variables, including 1) four variables—park size, the presence of an entrance fee, the presence of water, and the vegetation cover percentage—to describe park characteristics; 2) the number of nearby bus stops to quantify park accessibility; 3) three variables—population density, average housing price and number of parks nearby—to describe the physical and socioeconomic characteristics of park surroundings; and 4) the distance to the urban center to describe the park location (Table 2). We selected these predictor variables because previous studies have shown that these variables potentially affect the number of visits to urban parks (Cohen et al., 2010; Giles-Corti et al., 2005; Grow et al., 2008; Keeler et al., 2015; Sallis et al., 1990; Schipperijn et al., 2010; Wang et al., 2015; Zhang et al., 2015).

Park sizes were calculated based on the polygon layer of parks derived from the Google Earth imagery described above. We used the LandScan™ high-resolution population dataset to calculate the population density around the parks (Bright, Coleman, Rose, & Urban, 2011). The dataset of housing price contained 7832 housing projects in Beijing, which were obtained from Ganjiwang, a popular website providing housing sales/rental information (Beijing City Lab, 2013). The vegetation coverage of each park was calculated using a land cover map derived from 2.5 m ALOS image data (Qian, Zhou, Yu, & Pickett, 2015).

We first calculated the bivariate correlations between check-in visits and their potential influencing factors by including all 127 parks. We then calculated the bivariate correlations including only the community parks, neighborhood parks and natural parks, assuming these three types mainly serve local residents (N = 90). The second case excluded cultural relics parks, recreational parks and large urban parks because these parks were visited not only by nearby residents but also by citizens from all of Beijing City and even tourists from across the country. For example, a study based on social media data reported that the Beijing Olympic Forest Park attracted visitors from different districts of Beijing and that 22.2% of online posts were from non-Beijing residents (Wang, Jin, Liu, Li, & Zhang, 2018). Therefore, the influences of the surrounding attributes of these parks might differ from those of all parks in general.

We further used multiple linear regressions (MLR) to investigate the impacts of these factors on check-in visits and examine their relative importance in predicting check-in visits. We log-transformed the response variable because the check-in data were not normally distributed. We calculated the variance inflation factor (VIF) for each predictor variable and the mean VIF to test the multicollinearity among predictor variables, as some of them were correlated.

As with the bivariate correlation analysis, we developed two models to quantify the effects of the 9 predictors on check-in visits, as well as their relative predictive importance. The first model included all 127 parks, but the second model included only the community parks, neighborhood parks and natural parks, assuming that these three types serve mainly local residents. We conducted all statistical analyses using SPSS 19.0™.

Table 1
Park types and descriptions.

Park Type (N = 127)	Description
Recreational park (n = 13)	Recreational parks include children's park, zoo, botanical garden, and sports field, characterized by variable size and variable location
Cultural relics park (n = 16)	Cultural relics parks are ancient relics and sites that located in urban built-up area, featured by tourism value and education value
Large urban park (n = 8)	Large urban parks are public spaces serving a wide range of residents, characterized by adequate facilities, large size and various activities
Natural park (n = 14)	Natural parks are characterized by natural scenery and ecological value, located in the suburbs
Community park (n = 31)	Community parks are parks that located in residential areas, designed for local residents’ recreational purposes
Neighborhood park (n = 45)	Neighborhood parks are small residential green spaces that located in a certain residential district, serving a much smaller population compared with community parks

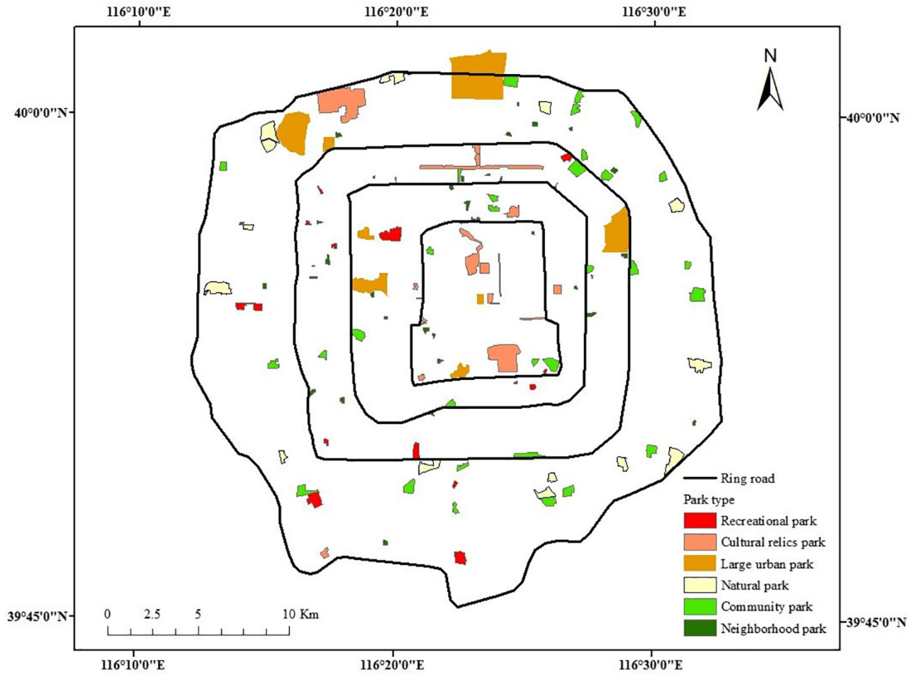

Fig. 1. The spatial distribution of different types of urban parks in the study area.

Table 2
Descriptions and data sources of selected predictors.

	Variables	Descriptions	Data Sources
Park attributes	Park size	Park size	Calculated from the polygon shape files
	Fee (1 = yes)	Entrance fee or not	Beijing Municipal Bureau of Landscape and Forestry
	PresWater (1 = yes)	Presence of water	Based on the classified ALOS image
	VegRate	Vegetation cover rate	Based on the classified ALOS image
Transport	N_Stops	Number of bus stops	Provided by Baidu Map, within the 500-m-buffer zone
Surrounding attributes	PopDensity	Population density	LandScan™ high-resolution population dataset, within the 1000-m-buffer zone
	Price	Average housing price	Beijing City Lab Data 8, within the 500-m-buffer zone
	N_Parks	Number of nearby parks	Calculated in ArcGIS 10.2, within the 1000-m-buffer zone
Location	Dis2UC	Distance to urban center	Calculated in ArcGIS 10.2, distance from park centroids to urban center

3. Results

3.1. Inequality in the distribution of park check-in visits

A total of 581,354 visits to urban parks were reported in Weibo from 2012 to 2016, with an average of 3767 visits per park, a standard deviation of 10,015, and a range from 16 to 51,651 (Fig. 2). The majority of these parks (n = 93) had 100 to 10,000 visits, while 19 parks had more than 100 visits and 15 parks (11.8% of the total) had more than 10,000 visits. These 15 parks had a total number of 450,996 visits, accounting for 74.6% of the total visits.

The number of park visits varied greatly by location (Fig. 2). The number of check-in visits tended to be greater in the northern area, especially for large parks such as the Olympic Forest Park, the Summer Palace and Chaoyang Park. Small and medium-sized parks received much less visits compared with large ones.

3.2. Differences in total check-in visits and visiting intensity among the 6 types of parks

The distribution of check-in visits and visiting intensity for all park types is shown in Fig. 3. The boxplot shows the median values, first and third quartiles, 95% confidence intervals of medians, and outliers.

The distribution of total visits per park showed large variability among park types. The number of visits tended to be high for cultural relics parks and large urban parks, low for natural parks (Fig. 3). The hypothesis of equal medians across groups was rejected (Kruskal-Wallis test, $p < 0.01$), and the Dunn test revealed that cultural relics parks and large urban parks hosted a significantly higher number of visits than any other types ($p < 0.01$). However, there was no significant difference between cultural relic parks and large urban parks.

Cultural relics parks had the largest number of visits per hectare, followed by large urban parks and neighborhood parks, while natural parks had the lowest number of visits per hectare. The hypothesis of equal medians among different types of parks was also rejected ($p < 0.01$). The Dunn test showed that cultural relics parks hosted a significantly greater number of visits per unit of area than recreational, community and natural parks ($p < 0.05$). However, there was no significant difference between cultural relics parks and neighborhood parks. While large urban parks had a larger number of visits per hectare than recreational, community and neighborhood parks (Fig. 3), the difference among these four types of parks was not statistically significant.

3.3. The effects of different factors on check-in visits and their relative importance

Table 3 shows the results from the Spearman's correlation analysis between the response variable and the selected predictor variables. For the case including all parks, the response variable was significantly

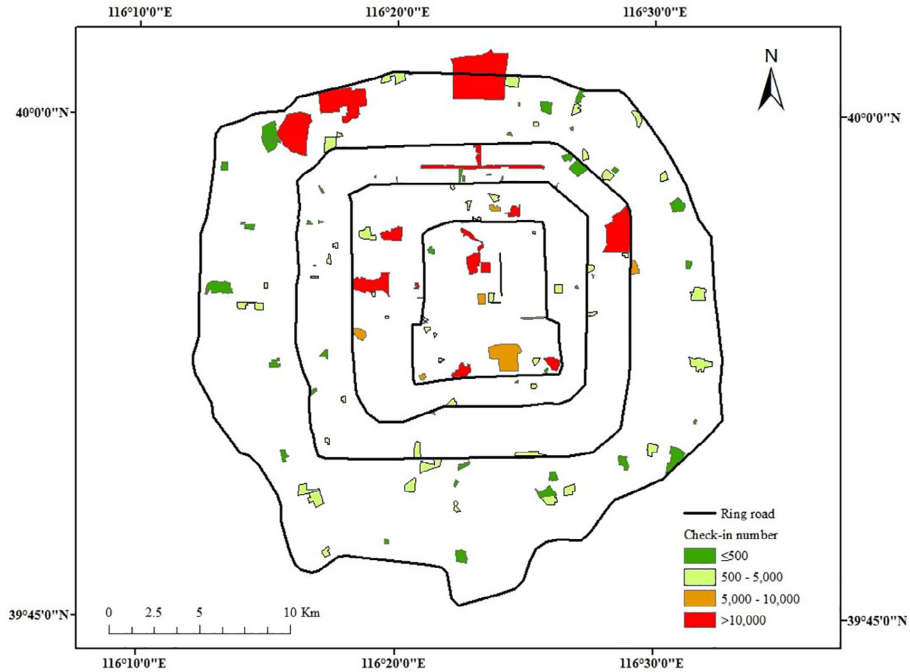

Fig. 2. Distribution of visits to Beijing parks measured by Weibo check-in numbers.

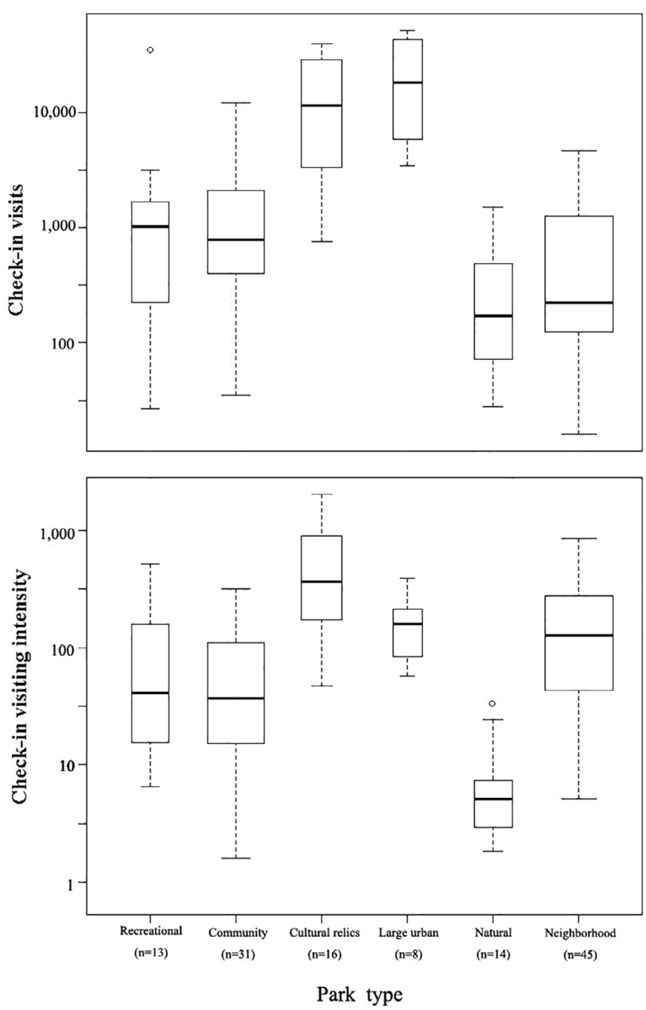

Fig. 3. Boxplot of visits (top) and visiting intensity (bottom) in the 127 parks of different types.

correlated with each of the predictors. When including only community parks, neighborhood parks and natural parks, the response variable was significantly correlated with four predictor variables: park size, the presence of an entrance fee, the number of nearby bus stops and distance to the urban center.

Table 4 shows the results of the two regression models: (1) the check-in numbers of all parks regressed against all 9 predictor variables (N = 127) and (2) the check-in numbers of parks, excluding recreational parks, cultural relics parks, and large urban parks, regressed against the 9 variables (N = 90). We included both the unstandardized regression coefficients and standardized coefficients. The standardized coefficients (beta coefficients) were used to determine the relative importance of the explanatory variables. The larger the absolute value of the standardized coefficient, the more important the variable. The results showed that the mean VIF were 1.9 and 2.1 for the two models, respectively, with VIF for most of the predictor variables less than 3, suggesting weak and ignorable multicollinearity among the predictor variables (Table 4).

For the model including all parks, the results of MLR analysis showed that four predictor variables, including two park characteristics variables (park size and the presence of an entrance fee), the number of bus stops, and distance to the urban center (Dis2UC), significantly affected the number of check-in visits (Table 4). Among these four factors, the presence of an entrance fee had the strongest influence, followed by park size, distance to the urban center, and the number of bus stops. Approximately 56.5% of the variation in check-in visits was explained by the model. Parks with an entrance fee received more visits than parks without a fee, presumably because these parks were well managed and had better facilities. Similarly, Ghermandi and Fichtman (2015) found that constructed treatment wetlands with entrance fees had a higher number of visits. Additionally, park size was positively related to check-in visits, indicating that parks with larger areas had greater numbers of visits. Distance to the urban center had a negative effect on check-in visits, indicating that parks farther from downtown received fewer visits. The number of nearby bus stops (N_Stops) was positively related to check-in visits, suggesting that increased accessibility via public transportation leads to more visits. Vegetation coverage and the presence of water had no significant effect. Characteristics of park surroundings also had no significant influence on check-in

Table 3
Spearman's correlations between check-in visits and the selected predictors in regression models.

	Park size	Fee	PresWater	Veg Cover	N_Stops	Pop Density	Price	N_Parks	Dis2UC
Model I	0.45**	0.55**	0.27**	−0.23*	0.52**	0.27**	0.25**	0.28**	−0.33**
Model II	0.22**	0.28**	0.12	0.04	0.25*	0.20	0.07	0.18	−0.29**

* Coefficient is significant at the 0.05 level.
** Coefficient is significant at the 0.01 level.

visits.

For community parks, neighborhood parks and natural parks, the MLR analysis showed that two park characteristics variables (park size and entrance fee) and distance to the urban center (Dis2UC) significantly affected the number of check-in visits (Table 4). After excluding recreational parks, cultural relics parks and large urban parks, model II explained approximately 26.7% of the variance. Among these three factors, Dis2UC had the strongest influence and had a negative effect on check-in visits, indicating that parks farther from downtown received fewer visits. Park size and the presence of an entrance fee remained significantly related to check-in visits, and both had positive effects on check-in visits. As in the previous model, the physical attributes of parks and characteristics of park surroundings showed no significance.

4. Discussion

In this study, we used check-in data from social media to quantify and compare the number of visits for different types of parks. We further investigated how different factors affected the use of parks in terms of check-in visits. The findings from this study provided explanations for the frequency of park use by urban residents at the city scale. The method presented here could be applied in other cities for comparison purposes.

4.1. Check-in visits varied among different types of urban parks

The results of this study show that although community parks and neighborhood parks have fewer visits than other park types, the visiting intensity of these parks is relatively high. This result raises a policy question regarding park size for park system planning processes: whether one large park or several small parks (i.e., community and neighborhood parks) will better meet residents' recreational demand in built-up urban areas; this warrants further research. This issue is similar to the single large or several small (SLOSS) debate in nature preserve design, that is, whether a single large nature preserve will contain more or fewer species than several small nature reserves (Burkey, 1989; Lindenmayer, Wood, Mcburney, Blair, & Banks, 2015; Rebelo & Siegfried, 1992; Tjørve, 2010). Our findings indicate that small parks with easy access and good management are important for nearby residents' entertainment and recreation. Several perceived benefits of small neighborhood green spaces were identified by a study in Latin America; these included providing places for women to relax and socialize and providing amenities for children to play (Wright Wendel et al., 2012). Grow et al. (2008) questioned whether closer small parks or more distant large parks had a stronger association with youth physical activity, and their findings supported a policy of building more small and accessible parks. However, a study in Odense found that small greenspaces were used less often and suggested that city planners find a balance by providing both smaller and larger urban greenspaces (Schipperijn et al., 2010). The roles of small parks with easy access and large urban parks are neither identical nor interchangeable, but rather overlapping and complementary (Liu et al., 2017). In the urban cores of cities such as Beijing, it is extremely unlikely that additional large parks will be established. However, there are many opportunities to build smaller community parks because many Chinese cities are still experiencing significant changes in land cover at a fine scale, even in urban cores, due to infilling development and/or city renewal (Qian et al., 2015; Zhou et al., 2018). Meanwhile, park managers could better protect existing urban community parks from development or the conversion of greenspace to buildings and infrastructure. Additionally, enhancing the quality of small parks can potentially provide more opportunities for residents' daily recreation (Haaland & van den Bosch, 2015).

4.2. Factors influencing park check-in visits and their relative importance

Our findings indicate that several factors explained the large variation in check-in visits across the 127 parks in Beijing. Park size had a significant positive relationship with check-in visits in both models, which is consistent with previous studies in other cities (Cohen et al., 2010; Giles-Corti et al., 2005; Schipperijn et al., 2010). Brown et al. (2014) found that park size was significantly related to the number of activities in parks. Large parks generally have more trees, walking

Table 4
Results from multiple linear regressions of park check-in visits on selected predictors.

Variable	Model I			Model II		
	Coefficient	Standardized coefficient	VIF	Coefficient	Standardized coefficient	VIF
Park size	$\mathbf{7.03} \times 10^{-3}$**	0.28	1.53	$\mathbf{2.84 \times 10^{-2}}$**	0.36	1.80
Fee	**1.65****	0.36	1.32	**1.16***	0.22	1.12
PresWater	0.45	0.12	1.44	0.19	0.06	1.42
VegCover	−0.20	−0.02	1.25	0.58	0.07	1.15
N_Stops	$\mathbf{7.83 \times 10^{-2}}$*	0.21	1.82	4.51×10^{-2}	0.09	1.74
PopDensity	-1.20×10^{-5}	−0.03	3.64	-3.49×10^{-5}	−0.13	4.48
Price	-3.87×10^{-6}	−0.02	1.68	-9.75×10^{-7}	−0.01	1.74
N_Parks	0.13	0.10	1.32	1.48×10^{-2}	0.01	1.42
Dis2UC	$\mathbf{-1.26 \times 10^{-4}}$*	−0.25	3.20	$\mathbf{-2.03 \times 10^{-4}}$*	−0.48	4.04
R^2	0.565			0.267		
Adjusted R^2	0.531			0.185		
mean VIF			1.91			2.10

* Coefficient is significant at the 0.05 level.
** Coefficient is significant at the 0.01 level.

paths, water features and birds, and thus people use the larger parks more often when they have both larger and smaller parks within a reasonable distance (Corti, Donovan, & Holman, 1996; Giles-Corti et al., 2005; Schipperijn et al., 2010). Our finding that parks charging admission fees had more visits is consistent with a previous study indicating that fee-based facilities scattered around the community were more effective in increasing exercise than public facilities (Sallis et al., 1990). Additionally, parks charging admission presumably organize more activities and have more attractive landscaping, as suggested by Cohen et al. (2010), which is likely to attract more people. Therefore, parks without a fee might attract more visitors by designing more trails and corridors, providing seats and shelters, and adding exercise equipment.

Distance to the city center was another factor that significantly affected the number of check-in visits in both models. Distance to the city center has a negative relationship with park check-in visits, suggesting that urban parks closer to the city center tended to have more visits, even after considering the effects of park size, accessibility (i.e., the availability of public transportation), etc. Our observation is that urban parks, especially community and neighborhood parks, in and close to the city center tend to have better design and maintenance and more amenities, which may attract more visits. Therefore, park managers and planners may want to allocate more resources to urban parks farther from the city center to improve maintenance and add amenities, and thereby increase residents' park use.

The predictive importance of the number of nearby bus stops differed for different park types. The number of nearby bus stops had a significant effect on check-in visits in the model for all park types, while there was no significance in the model for community parks, natural parks and neighborhood parks. We inferred that cultural relics parks, large urban parks and recreational parks were open not only to nearby residents but also to all citizens in Beijing, especially cultural relics parks that attracted tourists from all over the country, and thus, convenient transport increased visits to these parks. This finding is consistent with that of previous studies showing the strong association between active transport to recreation sites and frequent use by children and adolescents (Grow et al., 2008), and it supports the need for transport policies that facilitate public transport to increase park visits in Beijing. For community parks and neighborhood parks, however, it is likely that visitors were largely from local communities within walking distance and did not rely on public transit.

Several variables showed no significant effect on check-in visits. We did not find that the presence of water would increase check-in visits. The vegetation cover rate also had no significant influence on park visits. Because the vegetation cover rate was an indicator of the quantity of vegetation rather than the quality, it could not by itself encourage park use (Zhang et al., 2015). Population density had a positive relationship with check-in visits when using the Spearman correlation analysis in Model I (Table 3). This result is consistent with those from previous studies (Keeler et al., 2015). However, population density does not play a significant role in predicting park visits when controlling for the effects of other predictor variables using multiple linear regression, suggesting that the correlation between population density and check-in visits may be due to the correlations of population density with other predictor variables (Cohen et al., 2010).

4.3. Limitations and future research

This paper used geotagged check-in data from social media as a proxy to measure the number of visits to urban parks, using Beijing as a case study. In contrast to time- and labor-intensive surveys, this method is more time efficient and can provide good spatial coverage. However, this method also has its own limitations. Because we did not have access to observed visitation rates, we were unable to test whether there was a good correlation between the check-in data and observed visitation rates. The relationship between the Weibo check-in rate and real visitation might vary across parks. For example, visitors to cultural relics parks and large urban parks might be more likely to "check in" there, but less likely to do so at "not so popular" parks, that is, community and neighborhood parks. This potential difference among parks may contribute to uncertainty in the analysis of the effects of influencing factors. The reliability and usefulness of the Weibo check-in data will be further validated when visitor statistics are publicly disclosed by park authorities in the future. Additionally, Weibo users may not be representative of all park users in the study area. The sociodemographic characteristics, such as age, gender, income, and education level of Weibo users might differ from those of the whole population. Consequently, a park-use analysis based on Weibo data might not reflect the overall use of urban parks. Finally, the regression models we used did not deal with the problem of spatial autocorrelation.

Model I explained 56.5% of the total variation, and model II explained only 26.7% of the total variation, indicating that additional user-based factors might affect park visitation. Park visitation can be influenced by an individual's socioeconomic (e.g., income, occupation and education) and sociodemographic characteristics (e.g., age and sex), as well as psychological factors (e.g., lack of interest, perception of safety), as reviewed in the Introduction. The spatial variation of these user-based characteristics was a potential driving factor of visitation variance, but to our knowledge, the spatial quantification of these factors is a significant challenge at present. Unlike with visitor surveys, we were not able to obtain the sociodemographic characteristics of Weibo users, which hindered further study of the relationship between park visitation and user-based factors.

Further research can build on the method and results of this study to measure recreational visits to urban parks. For example, geotagged photograph data have been used to estimate the travel cost of lake destinations (Keeler et al., 2015). Similar methods could be used to assess the recreational benefits of parks if user-profile information containing the hometowns of Weibo users could be obtained in the future. Additionally, semantic analyses of social media content have been used to reveal landscape values (Chen et al., 2017) and to assess multiple cultural services on a regional scale (Martínez Pastur, Peri, Lencinas, García-Llorente, & Martín-López, 2015). Adopting similar methods, social media content posted by park users could be further interpreted to help understand visitors' emotions or feelings and to assess the multiple benefits provided by urban parks.

5. Conclusion

We used geotagged Weibo check-in data as an indicator of park visits within the 5th ring road of Beijing and investigated factors influencing the number of check-in visits. Using geotagged data from social media sources to estimate visitation improved geographical coverage compared with traditional research methods, such as visitor surveys and on-site counting. This study used geotagged check-in data as a proxy for visitation to urban parks, providing complementary information to explore residents' preferences related to the cultural ecosystem services of parks.

The results suggest that planners should pay more attention to the role of small neighborhood parks in urban greenspace systems. This study provides support for the recreational value of such parks by identifying their high visitation intensity. We found that park size, entrance fees, and distance to the city center significantly affect park use for all types of parks. Larger parks with entrance fees and closer proximity to the city center tend to have more visits. The number of bus stops was positively related to check-in visits, suggesting that increased accessibility via public transportation leads to more visits. However, for parks that serve mainly local residents, the number of bus stops does not play a significant role in park use. These findings increase the understanding of park use variations in Beijing city and identify park characteristics that facilitate the use of urban parks. We recommend that improving park maintenance and increasing transport accessibility are effective measures to encourage park use. The existence of well-managed, accessible, small green spaces in the park system is also essential to meeting local residents' recreational demand.

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